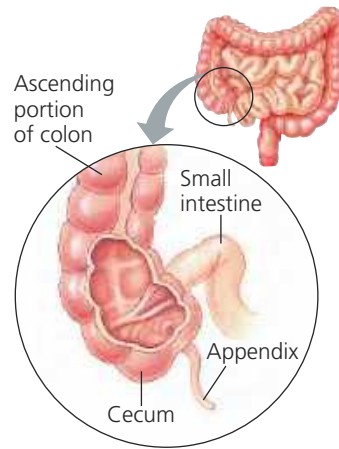


One arm of the T is the 1.5-m-long **colon**, which leads to the rectum and anus. The other arm is a pouch called the **cecum**. In animals that eat large amounts of plant material, the cecum has an important role in fermenting ingested material. In humans, the cecum is small and has an **appendix**, a finger-shaped extension that acts as a reservoir for symbiotic microorganisms, which are discussed in Concept 41.4.



▲ **Figure 41.14** Junction of the small and large intestines.

The colon completes the recovery of water that began in the small intestine. What remain are the **feces**, the wastes of the digestive system, which become increasingly solid as they are moved along the colon by peristalsis. It takes approximately 12–24 hours for material to travel the length of the colon. The undigested material in feces includes cellulose fiber. Although it provides no caloric value (energy) to humans, it helps move food along the alimentary canal.

If the lining of the colon is irritated—by a viral or bacterial infection, for instance—less water than normal may be reabsorbed, resulting in diarrhea. The opposite problem, constipation, occurs when the feces move along the colon too slowly. Too much water is reabsorbed, and the feces become compacted.

The community of bacteria living on unabsorbed organic material in the human colon contributes about one-third of the dry weight of feces. As by-products of their metabolism, many colon bacteria generate gases, including methane and hydrogen sulfide, the latter of which has an offensive odor. These gases and ingested air are expelled through the anus.

The terminal portion of the large intestine is the **rectum**, where the feces are stored before elimination. Two sphincters separate the rectum and the anus; the inner one is involuntary and the outer one is voluntary. Periodically, strong contractions of the colon create an urge to defecate. Because filling of the stomach triggers a reflex that increases the rate of contractions in the colon, the urge to defecate often follows a meal.

Having followed a meal through the alimentary canal, we'll look next at some adaptations of this general digestive plan in different animals.

CONCEPT CHECK 41.3

1. Explain why a proton pump inhibitor, such as the drug Prilosec, can relieve the symptoms of acid reflux.
2. The acids in bile salts have both lipid-soluble (hydrophobic) and water-soluble (hydrophilic) surfaces. How is such an organization beneficial for the role of bile salts in digestion?
3. **WHAT IF?** Predict what would happen if you mixed gastric juice with crushed food in a test tube.

For suggested answers, see Appendix A.

CONCEPT 41.4

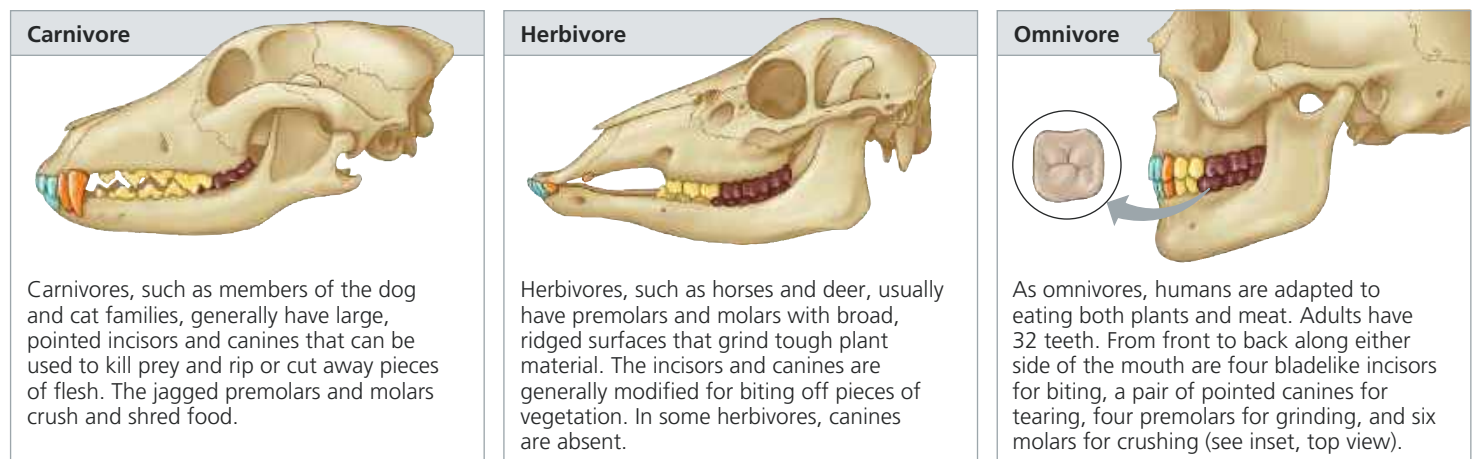
Evolutionary adaptations of vertebrate digestive systems correlate with diet

EVOLUTION The digestive systems of vertebrates are varied, with many adaptations linked to an animal's diet. To highlight how form fits function, we'll examine a few of them.

Dental Adaptations

Dentition, an animal's assortment of teeth, is one example of structural variation reflecting diet (**Figure 41.15**). The

▼ **Figure 41.15** Dentition and diet.



Key ■ Incisors ■ Canines ■ Premolars ■ Molars